

Mechanistic Insight Into Ionizable Cationic Lipid-Mediated Endosomal Escape via Transient Lipid Complexes

David Noel ZIMMER^{1,2}, Friederike SCHMID², Giovanni SETTANNI^{1,2}

¹ Faculty of Physics and Astronomy, Ruhr University Bochum, Bochum, Germany; ² Institute of Physics, Johannes-Gutenberg University Mainz, Mainz.

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ABSTRACT

RNA-based therapeutics have demonstrated remarkable efficacy and hold great promise for future applications in personalized medicine. The most common delivery systems for these drugs are lipid-based nanoparticles (LNPs), which incorporate ionizable cationic lipids (ICLs) as key components. Among other, ICLs are believed to facilitate endosomal escape of the cargo by interacting with anionic lipids in the endosomal membrane, although the underlying molecular mechanism remains unclear. One prevailing hypothesis suggests that membrane destabilization is mediated by cone-shaped complexes formed between ICLs and endosomal anionic lipids. However, no clear evidence of stable co-localization of anionic and cationic lipids has been presented so far. Here¹, to address this gap, equilibrium and nonequilibrium molecular dynamics simulations of model membrane systems containing DODMA (ICL), DOPS, or PI3P (anionic lipid) and DOPE or cholesterol (helper lipid) are performed. The results confirm the absence of co-localization at equilibrium but reveal transient formation of cone-shaped complexes during lamellar-to-inverted-hexagonal phase transitions, which considerably accelerates the transition process. These findings suggest that transient lipid–lipid interactions, rather than stable complexes, may play a critical role in facilitating endosomal escape. This mechanistic insight may inform the rational design of ICLs tailored to interact with specific endosomal anionic lipids, thereby enabling more effective and targeted delivery strategies for RNA-based therapeutics.

Fig. 1 Schematic drawing of the free energy barrier between the lamellar phase and the hexagonal phase in lipid bilayers and the associated transition state.

REFERENCES

1: D. N. Zimmer, F. Schmid, and G. Settanni, "Mechanistic Insight Into Ionizable Cationic Lipid-Mediated Endosomal Escape via Transient Lipid Complexes." *Small* 22, no. 32 (2026): e13399. <https://doi.org/10.1002/sml.202513399>